

GenAICHI: Generative AI and HCI

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ABSTRACT

This workshop applies human centered themes to a new and powerful technology, generative artificial intelligence (AI). Unlike AI systems that produce decisions or descriptions, generative AI systems can produce new and creative content that can include images, texts, music, video, and other forms of design. The results are often similar to results produced by humans. However, it is not yet clear how humans make sense of generative AI algorithms or their outcomes. It is also not yet clear how humans can control and more generally, interact with, these powerful capabilities. Finally, it is not clear what kinds of collaboration patterns will emerge when creative humans and creative technologies work together.

It is time to convene the interdisciplinary research domain of generative AI and HCI. Participation in this invitational workshop is open to seasoned scholars and early career researchers. We solicit descriptions of completed projects, works-in-progress, and provocations. Together we will develop theories and practices in this intriguing new domain.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Computing methodologies** → **Artificial intelligence**.

KEYWORDS

Generative AI; Design; Bias; Uncertainty.

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1 INTRODUCTION

Fooling Justin Bieber into picking a fight with Tom Cruise [32], completing Beethoven’s unfinished “Symphony No. 10” [39], and developing fake faces for stock photography [60] are some of the results that Generative Artificial Intelligence (AI) systems have recently brought to the world. What all of these examples have in common is that the AI does not simply categorize data and interpret text as determined by models, but instead creates something new—e.g., in images [28, 47], molecules [37, 56], or designs [38]. This work moves the purpose of AI systems from problem solving to problem finding, and it tends to change the “role” of the AI from decision-maker to human-supporter [48, 62]. In this workshop we focus on various aspects of generative AI and its interactions with humans, including the design of systems based on generative AI, ethical issues related to their design and use, and useful patterns for collaboration between humans and generative AI in different domains.

Generative AI can be defined as an AI system that uses existing media to create new, plausible media [37, 38, 46, 47, 56]. This scope is broad, and the generative potential of AI systems varies greatly. Over the last decade, we have seen a shift in methodology moving from expert systems based on patterns and heavy human curating [e.g., 16, 22, 40] towards stochastic and generative models such as Generative Adversarial Networks (GANs) that use big data to produce convincingly human-like results in various domains [14, 19, 47, 59].

In this workshop, we focus on the unique challenges that emerging generative AI methods pose to designers in various fields. These include, for example, the limited ability that designers and users have in gaining an understanding of the inner workings of these models: We cannot inspect the code or make sense of the latent vectors used in them and the generated results may vary. In addition the creative process is inherently fuzzy and involves elements of surprise and problem invention rather than achieving specific goals. Therefore we present that generative AI proposes new challenges for human-centered design due to the serendipitous and uncertain nature of the design space and especially for use cases where users work with the AI.

Drawing from the long tradition of computer supported collaborative work and creativity support systems, the idea of humans and AI agents working together to achieve creative results is becoming more and more commonplace, and a research paradigm of human-computer co-creativity and mixed-initiative creative interfaces is

emerging. In human-computer co-creativity, creative partners accept each others' suggestions into the space of possible creative products [51] and both the human and the computer are influenced by each others' contributions [7] culminating in sharing creative responsibility over the resulting product [22]. The new paradigm is characterized by the mixing of computer and human initiative [61] and human-computer co-creativity can be seen to flourish in the middle of a continuum between human creativity and autonomous computational creativity [10]. Emerging generative AI methods offer great opportunities for designing creative AI agents that are capable of generating useful artifacts with which they can contribute to the creative process of humans.

Despite powerful generative AI methods becoming more and more accessible for designers of creative systems, we still know relatively little about how to make generative AI methods interactive, how to design good user experiences around them and what kind of methods best facilitate design in this problem area defined by the open-endedness of creative work. Even less is known about the long term effects of the new technology to the creative practice of artists, designers and laypersons, the role generative AI based interactive systems will eventually take in society, and what kind of regulations will eventually govern the space of design in this area. We propose this workshop to unite a disparate community around generative AI to investigate questions related to the design, evaluation, deployment and ethics of interactive generative AI.

1.1 Previous Workshops

The general topic of AI has led to multiple workshops in the SIGCHI traditions [e.g., 2, 11, 25, 29, 33, 34]; however, workshops that specifically address *generative* AI and user experience [17, 18, 36, 52, 53] have not yet taken place at the CHI conference. With this submission, we propose to specialize the broad topic of generative AI to focus on the rich HCI possibilities of this emerging technology. Below, we describe some projects that have inspired our workshop.

2 BACKGROUND

2.1 Generative AI

Recent advances in generative AI methods show impressive abilities in diverse fields from images [4, 15, 42, 42] to text [41], music [14, 20], physical products [30, 31], code [59], and movement [57]. Whereas previous eras of generative AI are rule-based or built on relatively small models, new models are trained on massive internet-scale data and have untold abilities that we are only beginning to discover (see Figure 1). Moreover, these models (or versions of them) are open to the public without having to retrain them yourself [5, 9, 44, 55]. This gives many more people the ability to access powerful models. For example, there are thriving internet communities of artists collectively discovering text-to-image synthesis can be used in the artistic process [1, 6, 21, 26, 43, 54]. However, there are many unanswered questions about how and what AI can generate and how can and should people be involved in the process.

2.2 Designing for Creativity

Novelty and utility or value [45] sometimes accompanied by surprise [28] form three basic criteria for evaluating creativity. The

products of emerging generative AI methods can be said to be creative since they can be novel, valuable and surprising in a context of use. This means emerging generative AI methods have great potential for facilitating creative work in some form or another. We propose that creative work in various domains, such as music, art, writing, and design, offers important application areas for generative AI methods.

There are some special, domain independent considerations for facilitating interaction design for creativity. The open-ended nature of creative work challenges expectations designers have about designing software for traditional, productivity centered domains; the requirements of creative work are vague, the measures of success unclear and the behaviors of users can be unorthodox [49]. This open-ended nature of creative work raises questions about the roles that Generative AI can play in the creative process [24] and how we evaluate and compare the human-AI systems that are emerging from the design for creativity [23]. Users of creative software are not necessarily even working towards a specific goal, but the process of creation itself can be the goal of autotelic creativity [8]. Therefore combining powerful generative AI methods with users working and playing in creative domains opens up the space of design to unforeseen dimensions.

2.3 Imperfection in AI Outcomes for Human Use

These human-oriented and human-directed systems are just now becoming usable and useful. Initial findings suggest that, while perfect operation and outcomes would be preferable, for human creativity support even imperfect results can be better than no support at all, as studied in humans' use of generated documentation from source code [35, 58] and in translated source code from one programming language to another [59]. Software engineers both adopt and adapt (and occasionally replace) the outcomes of generative AI systems. The principle of AI-imperfection has not yet been parameterized: How bad does a generated outcome have to be, for a human to abandon it and re-do the work from scratch? How would domain experts negotiate the trade-offs in quality vs. productive-cost of generated outputs?

Many generative systems have been *productive* of new instances and new combinations learned from a class of exemplars or other strategies that make use of known histories of work [28, 38, 47, 56]. However, systems that create previously unknown instances that humans would consider to be *creative* (in terms of novelty, utility and surprise) are less understood. Deterding et al. [10] conducted a workshop in 2017 on interactive *creative interfaces*, and Spoto and Oleynik [50] created a *Library of Mixed Initiative Creative Interfaces* based on the materials in the workshop. Only a few of these experiments were specifically generative in their algorithms. Systems that are capable of producing novel instances are only recently documented [e.g., 3, 12, 13], and tooling that allows domain experts to control those novel or creative outcomes is even rarer [e.g., 27]. There are few formal human-centric, empirical evaluations of the quality of generated outcomes.

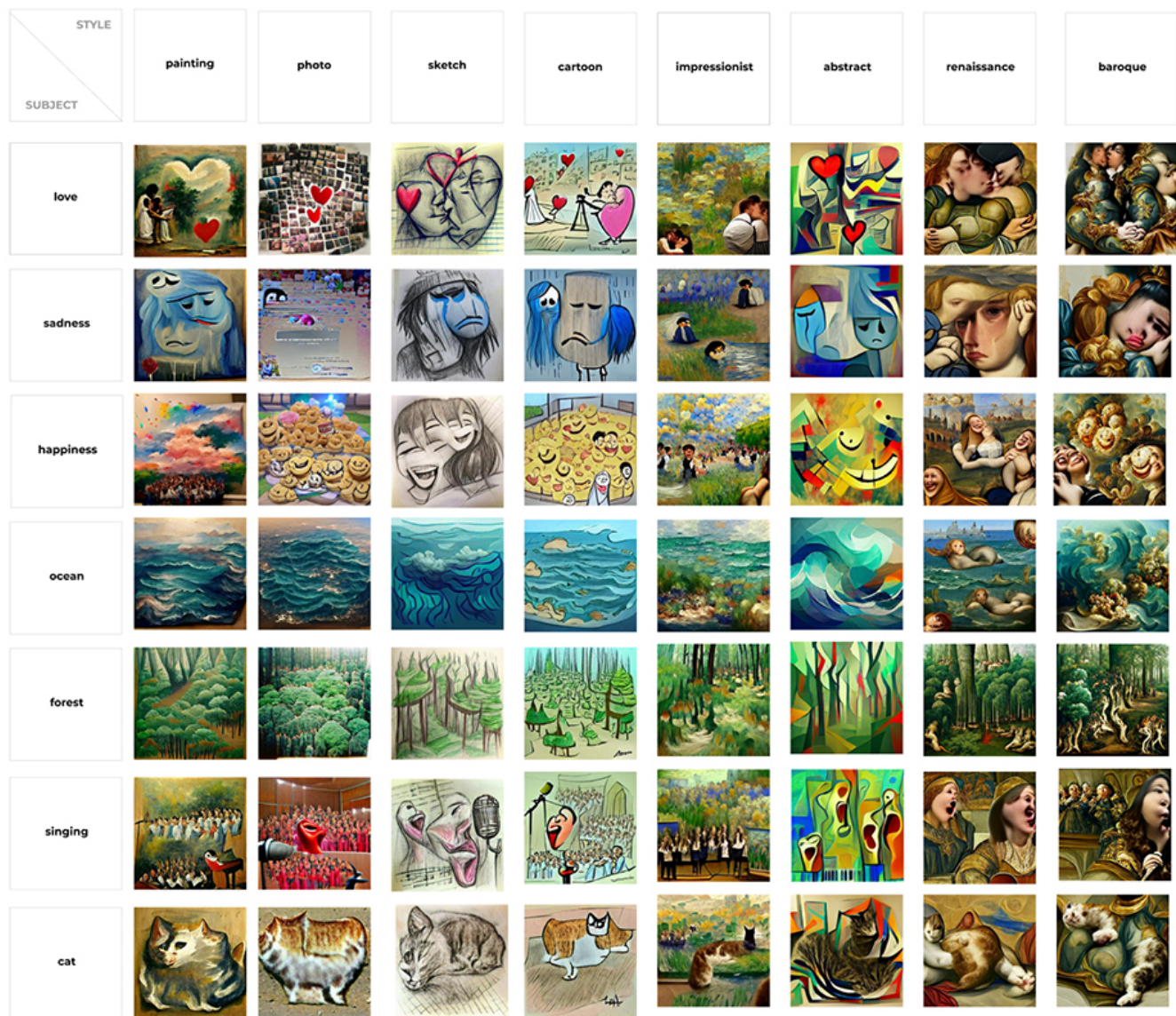


Figure 1: Images generated from text prompts in the VQGAN+CLIP model. Images for multiple subjects and various styles are shown to demonstrate the wide ability of image generation models trained with internet-scale data.

3 TOPICS AND THEMES

Our workshop is open to diverse interpretations of interactive generative AI, characterized by the AI systems' abilities to make new things, learn new things and foster serendipity and emergence. We are interested in several dimensions of generative AI, including mixed initiative, human-computer collaboration, or human-computer competition, with the main focus on interaction between humans and generative AI agents. We welcome researchers from various disciplines, inviting researchers from different creative domains including, but not limited to art, images, music, text, style transfer, text-to-image, programming, architecture, design, fashion and movement.

Because of the far-reaching implications of Generative AI, we propose the following list of non-exhaustive, thematic questions to guide our discussions at the workshop:

- What is generative AI and how can we leverage diverse definitions of it? Does generative AI go beyond intelligent interaction? What distinguishes generative AI?
- How do you design in this space? What design patterns do we need to think about? How does uncertainty play into this and how to we help people set expectations to designing with AI?

- Do generative AI algorithms contribute needed serendipity to the design process—or simply randomness—or worse, chaos?
- Is presenting AI as a desirable and “objective” method appropriate for generative AI?

We encourage people to write and answer their own questions as well. We hope that the workshop leads to new ways-of-thinking.

These themes can be addressed within the following topics:

- The emerging capabilities of generative AI.
- Generative AI existence in different domains including (but not limited to) images, music, text, design, and motion.
- The role of generative AI in assisting, replacing, and regenerating human work.
- Human-AI collaboration and co-creative systems.
- Ethical issues including misuses and abuses, provenance, copyright, bias, and diversity.
- The uncanny valley in Human-AI interactions.

As above, we encourage people to add new topics and domains.

4 WEBSITE

Our workshop website is: <https://generativeaiandhci.github.io> (currently online).

5 PRE-WORKSHOP PLANS

We will publicize the workshop via ACM, HCI, and AI distribution lists, plus social media [i.e., similarly to 2, 11, 25, 29, 33, 34]. We will also reach out to selected scholars to encourage them to submit their work.

We will request participants to submit completed work, works-in-progress, and provocations. Submissions should use the single-column ACM small format (see <https://chi2022.acm.org/for-authors/presenting/papers/chi-publication-formats/>). While the CHI 2022 instructions for *papers* say that there is no page-limit, we request that each submission be limited to four pages of content; references may appear on subsequent pages, and are not counted toward the four-page limit. The four-page submissions may take the form of reports, essays, or pictorials (for more information about pictorials, please see the section “What are Papers and Pictorials?” at <https://dis.acm.org/2021/papers-and-pictorials/>).

The co-organizers will select submissions for inclusion in the workshop. We anticipate a workshop size of 20-30 participants. Selection will be based on uniqueness and/or provocativeness of content, engagement with the themes and topics in the workshop call, and potential for contribution to this new community. All submissions will be subjected to single-blind peer-review by at least two experts. If we receive a large number of submissions, we will recruit a program committee from among known researchers and scholars who have worked at the intersection of HCI and generative AI.

6 WORKSHOP STRUCTURE

The goals of the workshop are to build community, exchange knowledge and to co-create new understandings in this emerging field. Therefore significant parts of the workshop will be reserved for networking and collaboration.

As we note below in Section 8, the workshop will run remote and it will be configured in multiple sessions to provide more inclusive access for participants in different timezones. We postpone setting up the exact schedules until we have a tentative list of participants, so that we can adjust the timing to fit their timezones.

The following list of topics describes the general content of our workshop sessions designed to reflect our goals to build a community, to exchange knowledge, and to co-create new understandings in this emerging field:

- (1) We will reserve significant time for the participants’ and organizers’ self-introductions and descriptions of research interests.
- (2) We will then invite a subset of the participants to present the most influential or promising work submitted to the workshop - either as individual talks or in a panel format. If we need to run multiple sessions for people from different geographies, then these presentations may need to take the form of pre-recorded videos and/or slide-decks with “scripts” that can be read-aloud by the workshop co-organizers.
- (3) We will preserve time for small-group discussions around themes of mutual interest, or generative HCI design challenges. These themes and challenges may be identified ahead of time, or they may be emergent from the presentations.

All participants’ submissions will be made available on our website before the workshop, and communications platforms will be arranged to enable asynchronous engagement before, during, and after our sessions. Online tools (e.g., Padlet) will be used to engage our community in collaborative ideation and critical discussion.

7 POST-WORKSHOP PLANS

We hope to build a persistent hybrid research area among HCI and generative AI. Toward that goal, we plan to submit an *Interactions* article or post for the general HCI audience.

In addition, we want to create resources for teaching and researching in this emergent domain. Therefore, we hope to publish a collection of papers from the workshop, organized as an invitational journal issue or a book of readings.

8 REMOTE/ONSITE PLANS

We assume that this workshop will be entirely virtual, so as to include people from around the world, and so as to avoid visa issues at the US border. To make a successful workshop under these circumstances, we propose to schedule several partial sessions, held so as to accommodate timezones of multiple continents. For example, if we receive submissions from Asia, Europe and Africa, and the Americas, then we would schedule three partial sessions (perhaps four hours per session) so as to include people from each of those regions. We will need to work adaptively and inclusively, depending on the submissions.

We realize that participants may not be able to attend all sessions, and we will therefore request that presenters prepare a video version of their presentation, and/or a slide-deck and a “script” that one of the organizers can read during the slide-deck. Each presentation will be followed by discussion. If authors can attend the discussion, that will make the discussion richer. However, we know from many

reading groups that it is possible to conduct a rich and productive discussion even if the author is not present.

One of our goals is to build community around the emerging hybrid field of HCI and generative AI. Therefore, we will begin each session with a period of introductions and interactions, followed by more formal presentations.

9 ORGANIZERS

Lydia B. Chilton is an Assistant Professor in the Computer Science Department at Columbia University. She is an early pioneer in decomposing complex tasks so that crowds and computers can solve them together. Her current research is in computational design—how computation and AI can help people with design, innovation and creative problem solving. Applications include: conveying a message within an image for journalism and advertising, developing technology for public libraries, improving risk communication during hurricanes, and helping scientists explain their work on Twitter.

Anna Kantosalo is a Postdoctoral Researcher at the University of Helsinki. The focus of her research is Human–Computer Co-Creativity and she is defining models and methods for building and describing systems in which humans and autonomous creative agents can work together. She has chaired the Future of Co-Creative Systems workshop adjoined with the International Conference on Computational Creativity twice.

Mary Lou Maher is a Professor in the Software and Information Systems Department at the University of North Carolina at Charlotte. Her early research in AI-based generative design has lead to a human centered approach to computational creativity and co-creative systems. She has Chaired the Creativity and Cognition Conference (2019) and the International Conference on Computational Creativity (2019) as well as organized several workshops on AI-based design and creativity.

Charles Martin is a Lecturer in Computer Science at the Australian National University. Charles works at the intersection of music, AI/ML and HCI. He studies how humans can interact creatively with intelligent computing systems and how such systems might fit in the real world. Charles has organized multiple generative-AI-focused workshops at the New Interfaces for Musical Expression conference.

Michael Muller is a Research Scientist at IBM Research in Cambridge MA USA. With colleagues, he has analyzed how domain experts make use of generative AI outcomes, and how humans intervene between "the data" and "the model" as aspects of responsible and accountable data science work. He has co-organized workshops on human centered data science at CHI, CSCW, and GROUP conferences, and a workshop on human centered AI at a NeurIPS conference.

Greg Walsh is an associate professor at the University of Baltimore where he teaches courses in Design. He is an interaction design researcher who focuses on user-centered, inclusive design for children and adults. His work seeks to include more voices in the design process and has been a recipient of a prestigious Google Faculty Research Award. His work has included participatory design sessions in Baltimore City libraries and is now exploring the use of generative AI as a co-design partner.

10 CALL FOR PARTICIPATION (250 WORDS)

This workshop applies human centered themes to a new and powerful technology, generative artificial intelligence (AI). Unlike AI systems that produce decisions or descriptions, generative AI systems produce new instances of types of data that can include images, texts, music, design, and motion. The results are often similar to results produced by humans.

However, it is not yet clear how humans can make sense of generative algorithms and outcomes. We have yet to understand what user interface technologies will enable humans to control, and more generally to interact with these powerful capabilities. These human-like capabilities put into question our current paradigms for mixed initiative user interfaces. Further, the unpredictability of "creative" algorithms raises new questions about how, when, and how much control humans may wish to share with these algorithms. Finally, it is not clear what kinds of collaboration patterns will emerge when creative humans and creative technologies work together.

In this one-day workshop, we will convene the interdisciplinary research domain of generative AI and HCI. Participation in this invitational workshop is open to seasoned scholars and early career researchers from diverse disciplines. We solicit descriptions of completed projects, works-in-progress, and provocations. Together we will develop theories and practices in this intriguing new domain. Please visit <https://generativeaiandhci.github.io/> to learn more about the workshop, and about how you can participate. Please send any questions to Michael Muller, michael_muller@us.ibm.com.

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